
Project Summary: G+10 Residential Building Schedule (Primavera P6)

Tool: Oracle Primavera P6 Professional 25 | **Author:** Kondepudi Sion Rani

1. Objective

To plan a complete construction schedule for a G+10 residential building (ground floor plus 10 floors) in Primavera P6, from site mobilization to final handover, and to check that the finish date is realistic for a building of this size.

2. Project at a glance

Item	Detail
Project ID	GT10BLDG
Start date / Data date	01-Oct-2026
Finish date	04-Feb-2028 (about 16 months)
Activities	140
Relationships	176 (FS with lags, SS)
Calendar	6-Day Week
Baseline	Baseline 1 - Original Plan (Initial Planning Baseline)

3. Scope and WBS

1. Preliminary & Mobilization: site survey, labor and equipment mobilization, site office and fencing
2. Substructure & Superstructure: excavation, anti-termite treatment, PCC, footing, backfilling, plinth beam, then columns and slab for ground floor to floor 10
3. Masonry & Plastering: block work and internal plastering per floor, external plastering
4. MEP Works: electrical conduiting, plumbing rough-in, HVAC ducting, fire fighting piping
5. Finishing Works: flooring, internal and external painting, doors and windows
6. Handover & Testing: MEP testing and commissioning, snagging and cleaning, final handover

4. How I built it

- Created the activity list with durations and the 6-day calendar, and defined logic in a relationship table (predecessor, successor, type, lag).
- Modeled a repeating floor cycle: column reinforcement, column concreting, slab shuttering (+1 day), slab reinforcement (start-to-start +2 days), slab concreting, de-shuttering (+7 days curing), then the next floor.
- Overlapped trades: block work starts 5 days after de-shuttering, plastering follows block work (+3 days), and flooring follows plastering (+5 days) and MEP rough-in.
- Saved the result as an initial planning baseline, and exported activities and relationships to Excel.
- Wrote a Python script (pandas, matplotlib) that generates a one-page summary and a full-project Gantt PDF from the P6 export.

5. Result

- The schedule runs 01-Oct-2026 to 04-Feb-2028 (about 16 months), which is within the 14-18 month range I would expect for a G+10 building.
- Superstructure is the longest phase, from December 2026 to November 2027. Finishing and handover follow it.

6. Known items still to fix

- SUP-F9-06 to SUP-F10-01 is Finish-to-Finish and should be Finish-to-Start.
- MAS-F10-01 to MAS-F10-02 has lag 0 instead of 3 days.
- The baseline should be updated after those fixes.
- MEP plumbing links are FS instead of SS (a simplification).

7. What I learned

- How constraints, Expected Finish dates and the data date change calculated dates.
- Why complete logic (no open ends) matters before trusting a finish date.
- How to check a schedule with a filter for blank predecessors and successors, and with the schedule log.
- How to manage baselines and export data for reporting.